



REQUIREMENTS DOCUMENT

EnerTrack

Energy Management for Multi-Site Construction Projects, Powered by AI

Intern Pre-Assignment — Phase 1: Planning & Requirements

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1. App Overview & Purpose

Problem

Construction companies typically run several sites in parallel. Each site consumes energy — electricity for machinery and lighting, gas or diesel for heating site cabins, and water for concrete work and cleaning. Today, this tracking is usually:

- Scattered across paper logs, spreadsheets, and individual site managers' memory.
- Reactive rather than proactive — overconsumption is only noticed once the invoice arrives.
- Invisible to head office — no consolidated, real-time view across all sites.
- A growing compliance burden, as CSRD/RSE reporting increasingly requires consolidated energy and carbon data.

Solution

EnerTrack is a centralized Energy Management platform dedicated to construction sites. Site managers log consumption readings (electricity, gas, fuel, water); the platform compares each reading against the site's own history, a similar site, or a defined budget/target; and an AI layer turns the resulting figures into a plain-language explanation of what changed and why. The result: energy waste is caught earlier, is easier to understand, and is easier to report on.

Differentiation

Energy-tracking tools already exist for construction (e.g. Citron[®], used by groups such as Demathieu Bard, for multi-site energy consumption monitoring). EnerTrack's angle is not “yet another consumption dashboard”, but the combination of: comparison logic (history / similar site / target) + AI-generated, human-readable explanations of anomalies — aimed at being usable without an energy-management background.

2. Target Users

- Site Manager — logs energy readings for their own construction site on a regular basis.
- Energy Manager — oversees energy performance across all sites, monitors alerts, prepares group-level analysis.
- Direction / Client — consults consolidated results and RSE/CSR reporting, without editing data.

3. Feature List (Prioritized)

Feature	Priority	Description
User authentication (sign-up / login / logout)	Must-have	Email/password auth via Supabase Auth (password reset is out of MVP scope).
Role-based access control (3 roles)	Must-have	Site Manager / Energy Manager / Direction, each with distinct permissions.
Site management (CRUD)	Must-have	Create, view, edit, archive construction sites.
Consumption reading entry (CRUD)	Must-have	Manual entry of electricity / gas / fuel / water readings per site and date.
Consumption dashboard	Must-have	Charts of consumption trends per site,

Feature	Priority	Description
		per energy type.
AI-generated anomaly explanation	Must-have	LLM call comparing current readings to history/target and generating a plain-language summary.
Site ranking & alerts	Should-have	Ranks sites by consumption; flags sites exceeding a threshold.
CO ₂ estimate	Should-have	Approximate CO ₂ equivalent based on consumption and energy type.
PDF/Excel export	Should-have	Export consolidated reports for RSE/CSR use.
Responsive UI	Must-have	Usable on desktop and tablet (site managers may use a tablet on-site).
CSV import of readings	Nice-to-have	Bulk import instead of manual entry.
Streaming AI responses	Nice-to-have	Bonus per assignment brief.

Example AI Prompt (anomaly explanation)

Illustrative prompt sent to the LLM when a user clicks “Generate AI summary” for a site:

“Here is the consumption data for site [Site Name]:
– Last month: 12,000 kWh (electricity), 3 m³ (water)
– This month: 15,000 kWh (electricity), 2.5 m³ (water)
– Monthly target: 13,000 kWh.
Write a 2–3 sentence summary explaining the situation and suggesting a likely cause.”

The response is displayed as a short text panel next to the site's consumption chart. Prompt wording will be refined during Phase 3 implementation.

4. Role List & Permissions

Note: the assignment brief uses admin/editor/viewer as reference labels. In EnerTrack these map to Direction (admin), Energy Manager (an intermediate elevated role with broad read access but no user management), and Site Manager (editor, scoped to one site).

Role	Description	Key Permissions	Site Access	Can Create Site?
Site Manager	Logs consumption data for their own assigned site. Cannot see other sites or manage users. MVP rule: exactly one Site Manager per site (1:1) — so reading ownership by site_id and by created_by are equivalent for this role.	Create/edit/delete own site's readings • View own site dashboard • Request AI summary for own site • Edit own profile	One site (own)	No

Role	Description	Key Permissions	Site Access	Can Create Site?
Energy Manager	Oversees energy performance across the whole company. Broad read access and export rights, but no account/site creation — an elevated reporting role rather than a full editor.	View all sites and readings • View rankings & alerts • Generate AI summaries for any site (stored in ai_summaries) • Export reports • Cannot manage user accounts or create sites	All sites	No
Direction (Admin)	Full system access, including user and role management. The first Direction account is created manually (Supabase seed / SQL) before any sign-up flow runs.	All Energy Manager permissions • Create/manage user accounts • Assign or change roles • Assign a Site Manager to a site • Create/archive sites • Delete accounts	All sites	Yes

Default Role & Sign-up Behavior

- New users sign up with the default role site_manager and no site assigned (site_id = null).
- Until Direction assigns them to a site, they land on a “Pending assignment” screen (“Your account has been created. Waiting for a site assignment by an administrator.”) rather than an empty/functional dashboard — this avoids ambiguity about whether the app is broken.
- Direction can change any user's role and site assignment at any time from User & Role Management (Screen 6).
- The very first Direction account is not created through the public sign-up form; it is seeded directly in Supabase before the app goes live, so there is always at least one admin able to assign roles.

Fourth Role (Client) — Decision

For the MVP, the read-only “Client” view described in Section 2 is merged into the Direction role (Direction dashboards are read-and-write; a client-facing export can reuse the same read queries). A dedicated Client role with its own RLS policy is a candidate extension from Phase 2 onward if time allows, but is not required to meet the 3-role minimum, which is already satisfied by Site Manager / Energy Manager / Direction.

Data Model Preview (high-level)

A detailed schema and RLS design belong to Phase 2, but the following entities are anticipated so the role/permission logic above is implementable:

- profiles — extends Supabase auth.users; stores role (site_manager / energy_manager / direction) and, for a Site Manager, their assigned site_id (nullable until Direction assigns one).
- sites — id, name, location, monthly energy budget, status (active / archived), created_by.
- readings — id, site_id, energy_type (electricity / gas / fuel / water), value, date, created_by. Since each site has exactly one Site Manager, site_id and created_by resolve to the same access rule for this MVP.

- ai_summaries — id, site_id, generated_text, created_by, created_at. Generated summaries are persisted so Energy Manager/Direction can review history and include them in exports, rather than being shown only once and discarded.

This is what an RLS policy would key off in Phase 2: a Site Manager's queries are restricted to rows where readings.site_id matches their profile's assigned site_id; a profile with site_id = null matches no rows, which is exactly the "pending" state described above.

5. Screen List / Wireframes

- Login / Sign-up — email + password.
- Site list (Direction / Energy Manager view) — cards per site: name, status badge (normal/alert), mini trend chart, "+ New site" action.
- My Site (Site Manager view) — reading entry form (energy type, value, date) + history chart, scoped to their one site.
- Site detail dashboard — consumption charts by energy type, comparison to history/target, "Generate AI summary" button and result panel.
- Consolidated dashboard (Direction) — cross-site ranking, alerts list, CO₂ estimate, export to PDF/Excel.
- User & role management (Direction only) — list of accounts, role assignment, site assignment for Site Managers.
- Pending assignment — message shown to a new Site Manager until Direction assigns a site (wireframe in the Design Document).

Low-fidelity wireframes for the first six screens are provided in Appendix A; the Pending screen wireframe is in the Design Document.

6. Technology Choices & Rationale

Layer	Choice	Rationale
Frontend & Backend	Next.js (App Router)	Single project for UI and API routes; file-based routing keeps the site structure simple to extend as features grow.
Database & Auth	Supabase (PostgreSQL)	Managed Postgres with built-in Auth and Row Level Security — lets access rules (e.g. a Site Manager only reading their own site's rows) be enforced at the database level, not just in the UI.
LLM Integration	Anthropic or OpenAI API	Used for the anomaly-explanation feature; exact provider to be confirmed based on available free-tier credits.
Styling	Tailwind CSS	Fast to iterate with AI coding tools; no separate CSS files to maintain.
Hosting	Vercel	Native fit for Next.js; automatic deployment on every push to GitHub.
Source Control	GitHub	Version history, small frequent commits per feature.
AI Dev Tools	Cursor / Claude Code	Used throughout all phases per assignment guidelines; all generated code is reviewed and understood before committing.

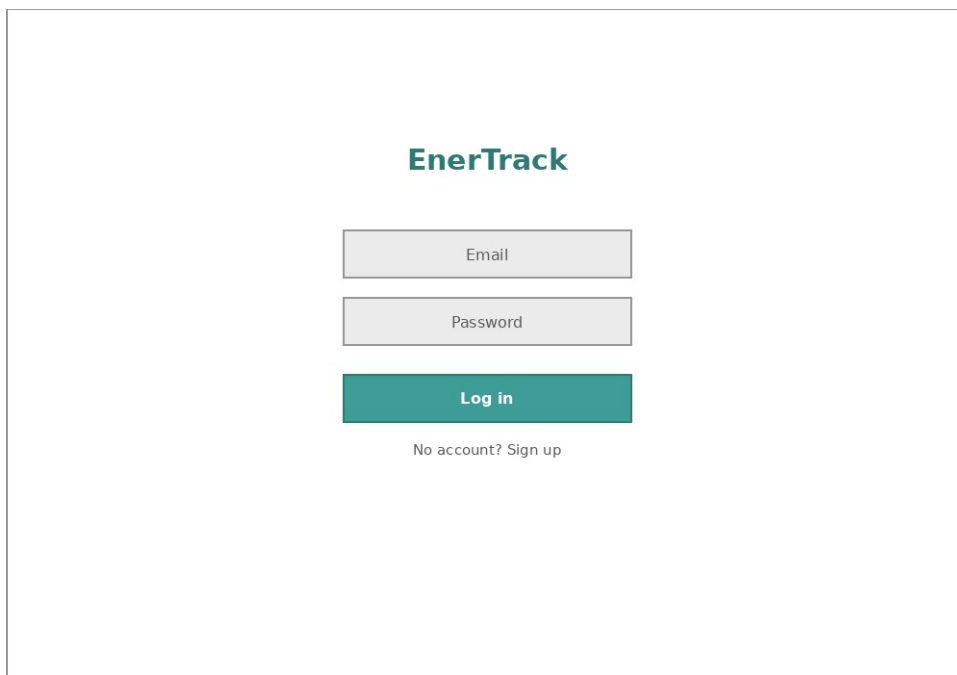
Scope Note

For this MVP, readings are entered manually rather than pulled from IoT meters. The document therefore describes the product as offering “centralized tracking, updated with every reading” rather than “real-time monitoring”. Automated/simulated data feeds are a possible extension for the demo if time allows.

Appendix A: Wireframes

Low-fidelity wireframes for the first six screens listed in Section 5. The Pending assignment screen is wireframed in the Design Document.

1. Login / Sign-up



EnerTrack

Email

Password

Log in

No account? Sign up

2. Site List (Direction / Energy Manager)

EnerTrack

Direction / Energy Manager

+ New site

Site A (Alert)

[trend chart]

Site B (Normal)

[trend chart]

Site C (Normal)

[trend chart]

3. My Site (Site Manager)

EnerTrack

Site Manager - My Site

New reading

Energy type (dropdown)

Value

Date

Submit

[history chart of readings]

4. Site Detail Dashboard

EnerTrack
Site Detail - Site A

[consumption charts: electricity / gas / water, vs history & target]

Generate AI summary

AI summary result panel (text)

5. Consolidated Dashboard (Direction)

EnerTrack
Direction - Consolidated Das

[site ranking by consumption]

[alerts list]

[CO2 estimate]

Export PDF / Excel

6. User & Role Management

EnerTrack
Direction - User & Role Mana

+ Invite user

User 1 - Site Manager - Site A

Change role

User 2 - Energy Manager - All sites

Change role

User 3 - Site Manager - Site B

Change role